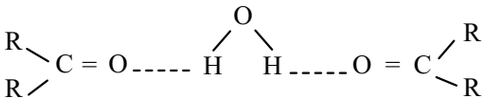
3. Solubility

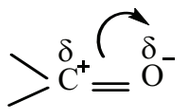
Lower members of aldehydes and ketones like formaldehyde, acetaldehyde and acetone are miscible with water in all proportions because they form hydrogen bond with water.

Solubility of aldehydes and ketones decreases rapidly on increasing the length of alkyl chain.



4. Dipolemoment:

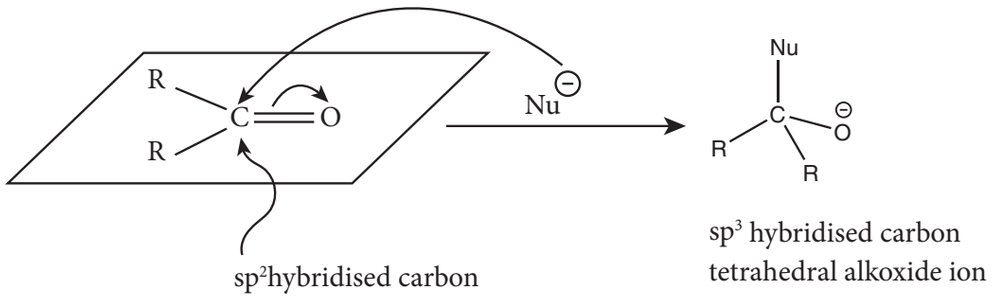
The carbonyl group of aldehydes and ketones contains a double bond between carbon and oxygen. Oxygen is more electronegative than carbon and it attracts the shared pair of electron which makes the carbonyl group as polar and hence aldehydes and ketones have high dipole moments.



12.5 Chemical properties of aldehydes and ketones

A) Nucleophilic addition reactions

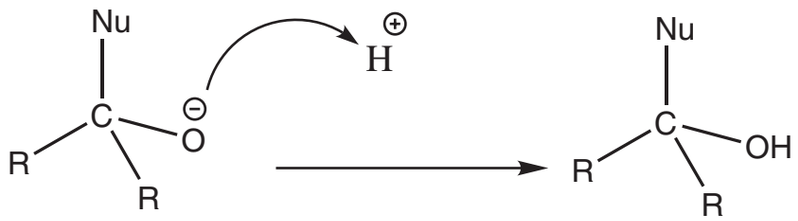
This reaction is the most common reactions of aldehydes and ketones. The carbonyl carbon carries a small degree of positive charge. Nucleophile such as CN^- can attack the carbonyl carbon and uses its lone pair to form a new carbon – nucleophile 'σ' bond, at the same time two electrons from the carbon – oxygen double bond move to the most electronegative oxygen atom. This results in the formation of an alkoxide ion. In this process, the hybridisation of carbon changes from sp^2 to sp^3.



sp^2hybridised carbon

sp^3 hybridised carbon
tetrahedral alkoxide ion

The tetrahedral intermediate can be protonated by water or an acid to form an alcohol.



In general, aldehydes are more reactive than ketones towards nucleophilic addition reactions due to +I and steric effect of alkyl groups.

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